Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/29/2016
PI Name:	Jaing, Crystal Ph.D.		
Project Title:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) Project		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jaing2@llnl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-424-6574
Organization Name:	Lawrence Livermore National Laboratory		
PI Address 1:	Physical & Life Sciences Directorate		
PI Address 2:	7000 East Ave,Mail Code L-452		
PI Web Page:			
City:	Livermore	**State:**	CA
Zip Code:	94550-9698	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/01/2015	**End Date:**	04/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
Contact Email:	elizabeth.taylor-23@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Feb 2016: No changes.		
COI Name (Institution):	Mehta, Satish Ph.D. (Wyle Laboratories, Inc.) Pierson, Duane Ph.D. (NASA Johnson Space Center) Smith, David Ph.D. (NASA Kennedy Space Center) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NNX15AJ29G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with various ISS modules using traditional culture-based methods and state-of-the art molecular techniques. The objectives of the ISS-MOP project are to measure presence of viral and select bacterial and fungal pathogens and correlate their presence on crew. The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. The hypothesis of the ISS-MOP project is that certain short-living viruses associated with inert environmental spaces in the closed habitat of the ISS can have an adverse influence on crew health. To test this hypothesis the team will catalog viruses associated with surfaces and air of the ISS modules using the all-encompassing Lawrence Livermore Microbial Detection Array (LLMDA), which contains probes to detect not only viruses but also clinically important human bacterial and fungal pathogens. In addition, skin and saliva samples from crewmembers will be requested to test the hypothesis. There are three aims proposed in this ISS-MOP project. While the cultivation-based assay (Aim 1) provides a valid estimation of microbial presence, it offers only a limited assessment of the phylogenetic/pathogenic diversity and physiological breadth of the microbial population present. Therefore, the team will use the novel and comprehensive LLMDA microarray, which is specifically designed to elucidate the pathogenic viral and microbial diversity profiles of complex samples to at least the species level (Aim 2). In addition, we will characterize the virulence and antibiotic resistance mechanisms associated with these pathogens from the ISS by developing a new Virulence and Antibiotic Resistance gene array and analyze the samples on this array. The final deliverable of the proposed project to NASA will be the ISS-MOP dataset, a compilation of all of the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat (Aim 3). The objectives of the ISS-MOP project directly respond to the National Research Council Committee for the Decadal Survey on Biological and Physical Sciences in Space recommendation P1 where it states, “to establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.” The ISS-MOP project will also provide answers to NASA Space Biology Guiding Questions: How do alterations in gravity affect in microbial cells (CMM1)?; Does the enhanced virulence or other genetic changes observed with microbial cultures flown in space occur in other species (CMM4)?; and What are the genetic changes underlying physiological responses to space flight demonstrated in biological systems (CMM5)?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with crew members and various ISS modules using both traditional culture-based methods and state-of-the art molecular techniques to a degree that has not previously been possible.
Task Progress:	• NASA grant to PI and co-I institutions were established during the first year of this project. Partial year 1 funding was awarded to PI and co-I institutions. • PI and co-I attended the American Society for Gravitational and Space Research (ASGSR) conference held in Alexandria, VA in November of 2015 and presented the technologies to be used in the ISS-MOP project. • Flight definition work has started to evaluate the air samplers to be used inflight to collect air samples for microbial characterization. The yield of cell mass collected will be compared between two samplers. Different types of filters are being evaluated to choose the filters with the least background microbial DNA. Team is making preparation to receive preflight samples in fall 2016 and inflight samples in spring 2017 under Increment 51/52. • Team prepared Institutional Review Board (IRB) documents for the study on environmental and crew sample microbiome. The IRB was approved. Team also presented at Informed Consent Briefing to present the project goals to the crew.
Bibliography Type:	Description: (Last Updated: 04/07/2023)
Abstracts for Journals and Proceedings	Jaing CJ, Smith DJ, Mehta S, Venkateswaran K. "Microbial Observatory of Pathogens in International Space Station: Bacteria, Fungi, and Viruses." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. https://asgsr.org/index.php/2015-abstracts , Nov-2015